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Topic: Driving Condition Classification

Dataset: BDD100K

FINAL PROJECT

Road/Visibility Classification from Dashcam images



DATASET

BDD100K is a large scale driving video dataset, which comprises different scenes types, such as city, street, residential areas and highways.

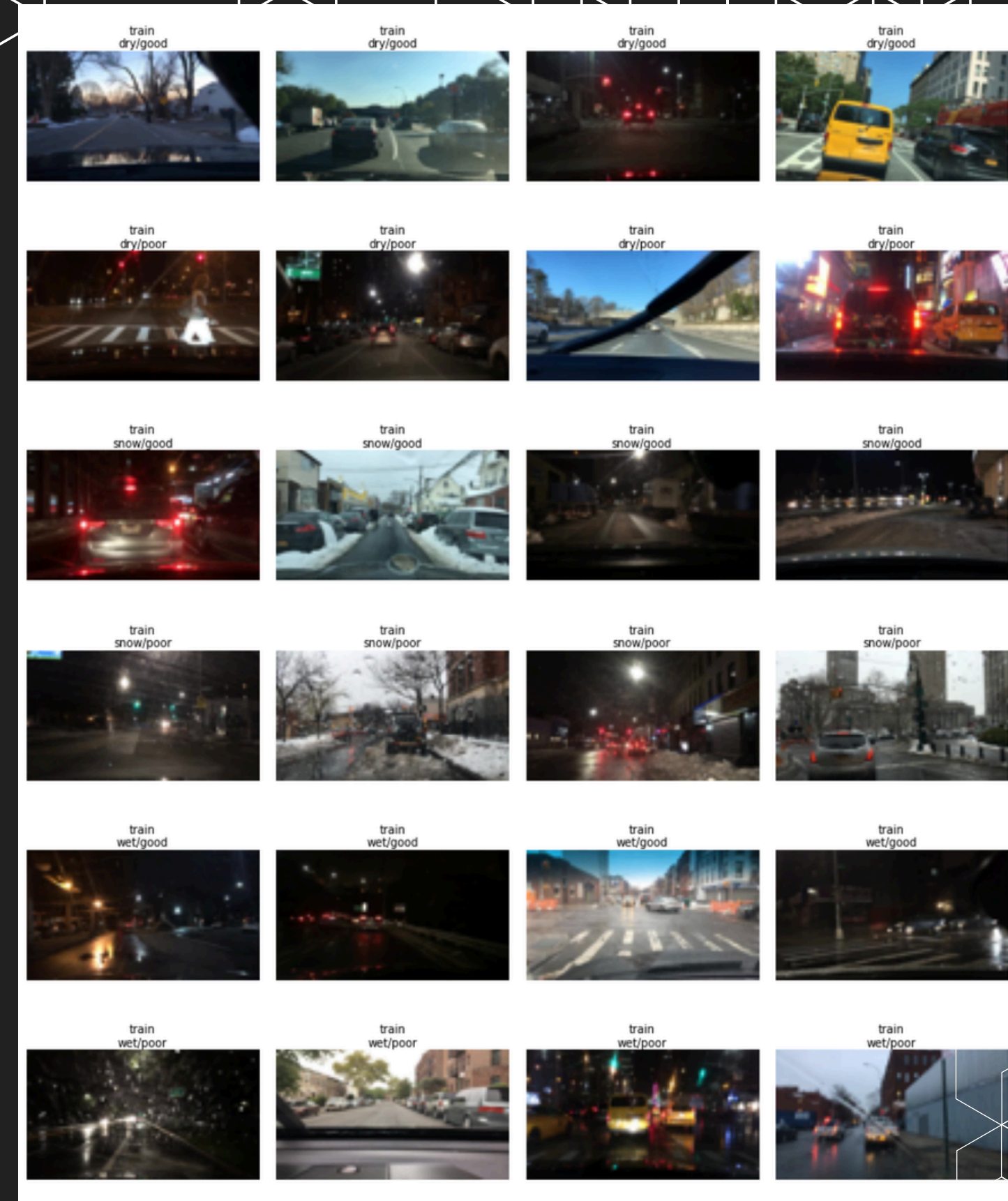
- has been captured under various weather conditions and time of the day
- 100K images manually annotated for a number of different tasks, including object detection or image segmentation
- each annotated image includes a label about the atmospheric weather condition (snowy, rainy, clear)
- this repository has annotated this further into road and visibility conditions

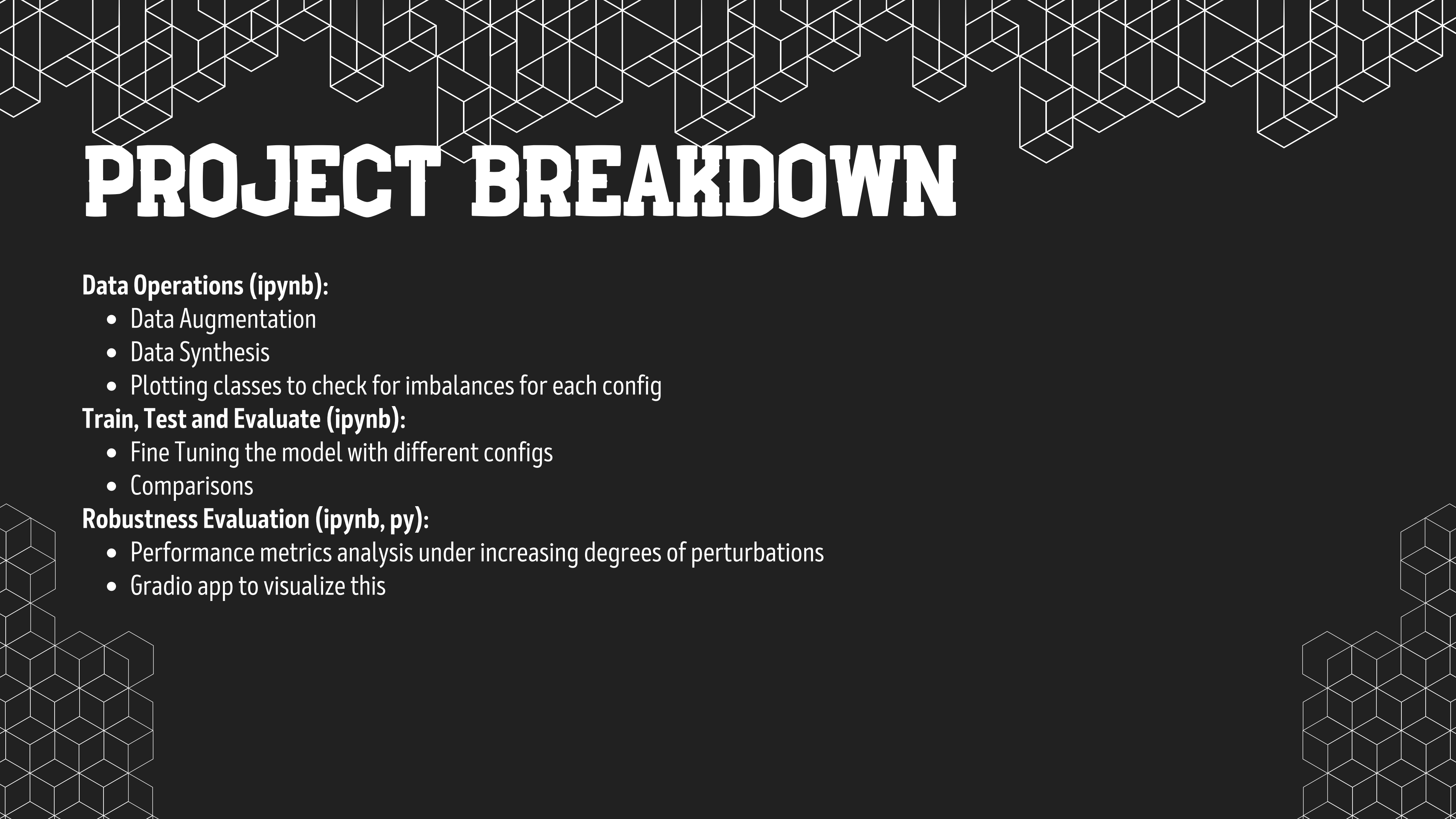
Classes:

Visibility: Good/Poor

Road Condition: Dry/Wet/Snow

Bukets = 6





PROJECT BREAKDOWN

Data Operations (ipynb):

- Data Augmentation
- Data Synthesis
- Plotting classes to check for imbalances for each config

Train, Test and Evaluate (ipynb):

- Fine Tuning the model with different configs
- Comparisons

Robustness Evaluation (ipynb, py):

- Performance metrics analysis under increasing degrees of perturbations
- Gradio app to visualize this

DATASET ANALYSIS + OPS



Augmentation

- Resized Crop
- Horizontal Flip
- Rotation
- Brightness variations
- Contrast variations
- Gaussian blur

At least 20 images per bucket

DATASET ANALYSIS + OPS



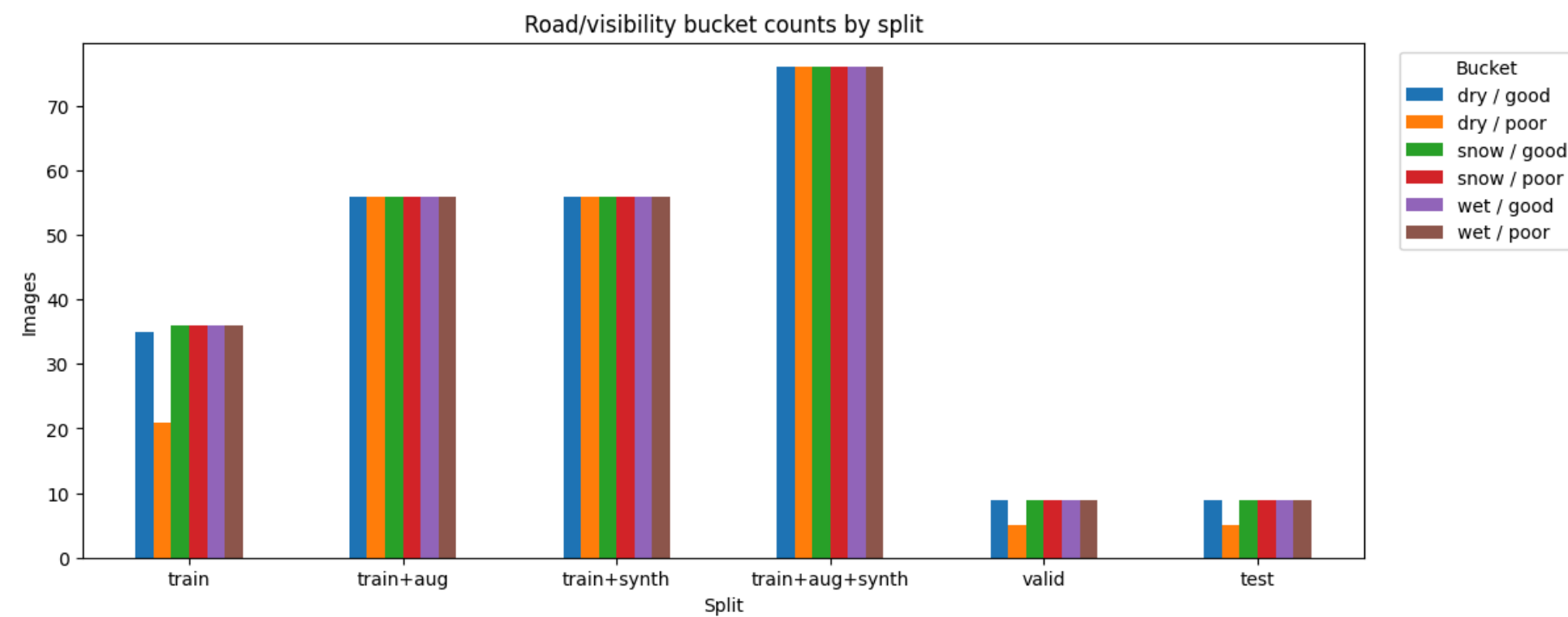
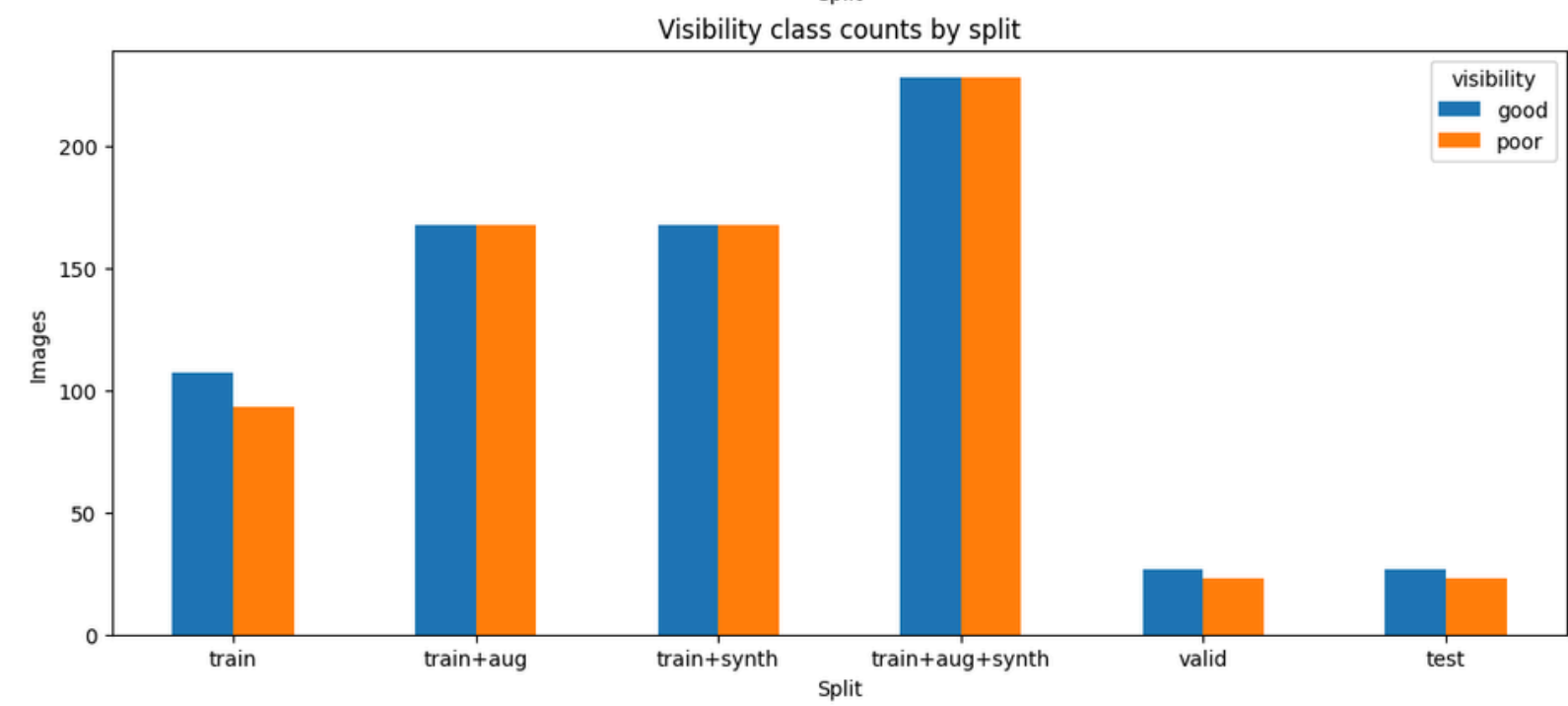
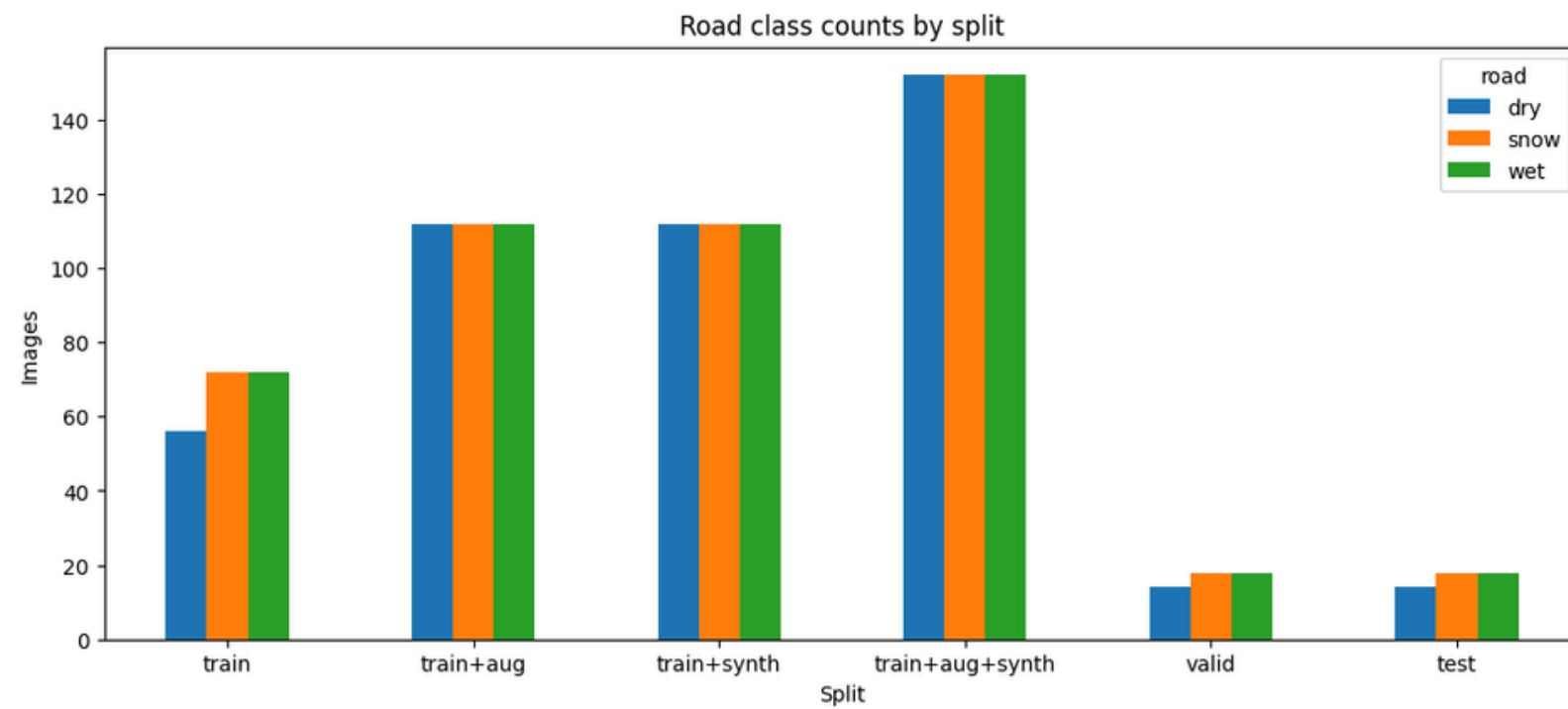
Synthesis

- Model: gpt-image-1-mini
- Shape: 1024x1024

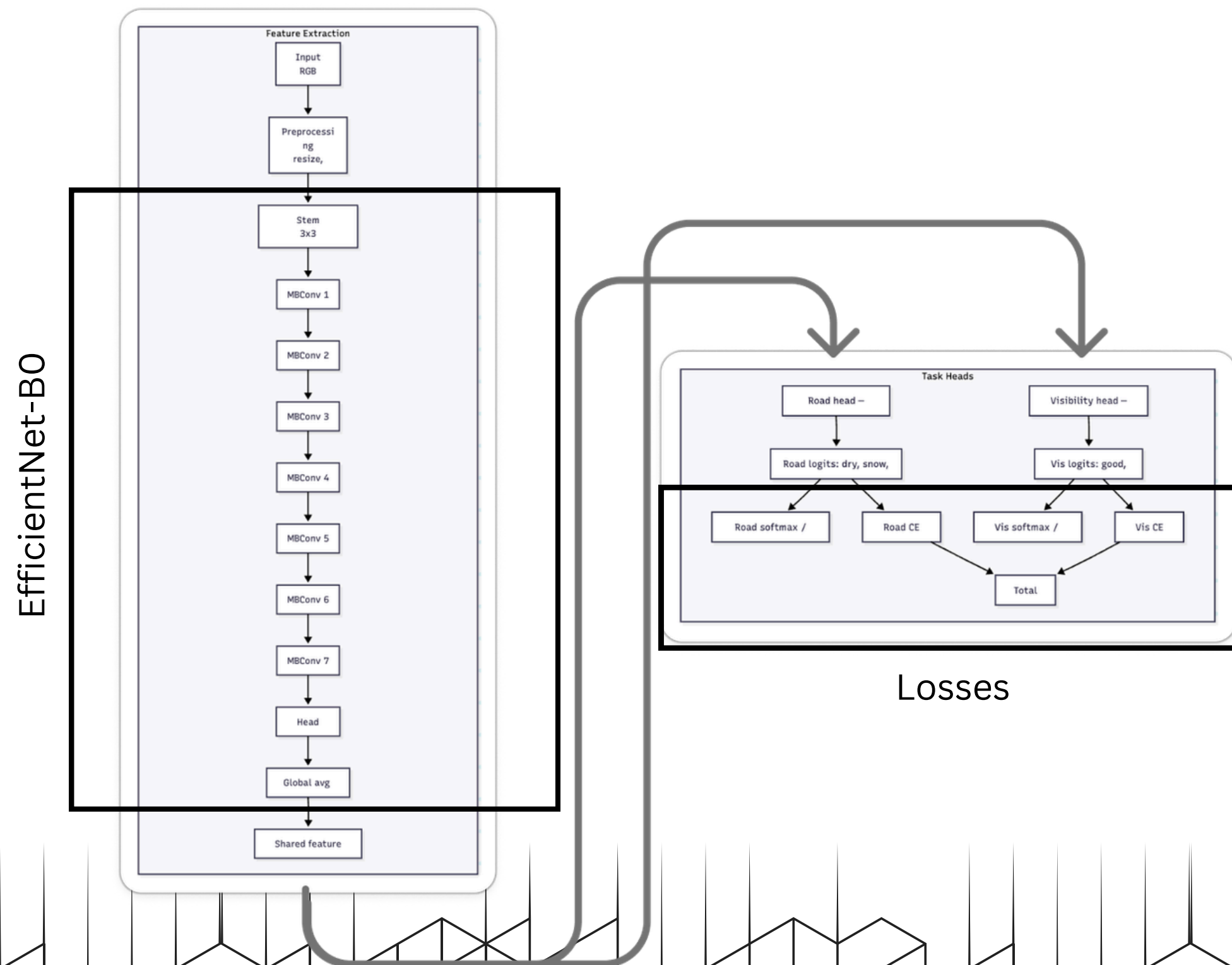
At least 20 images per bucket



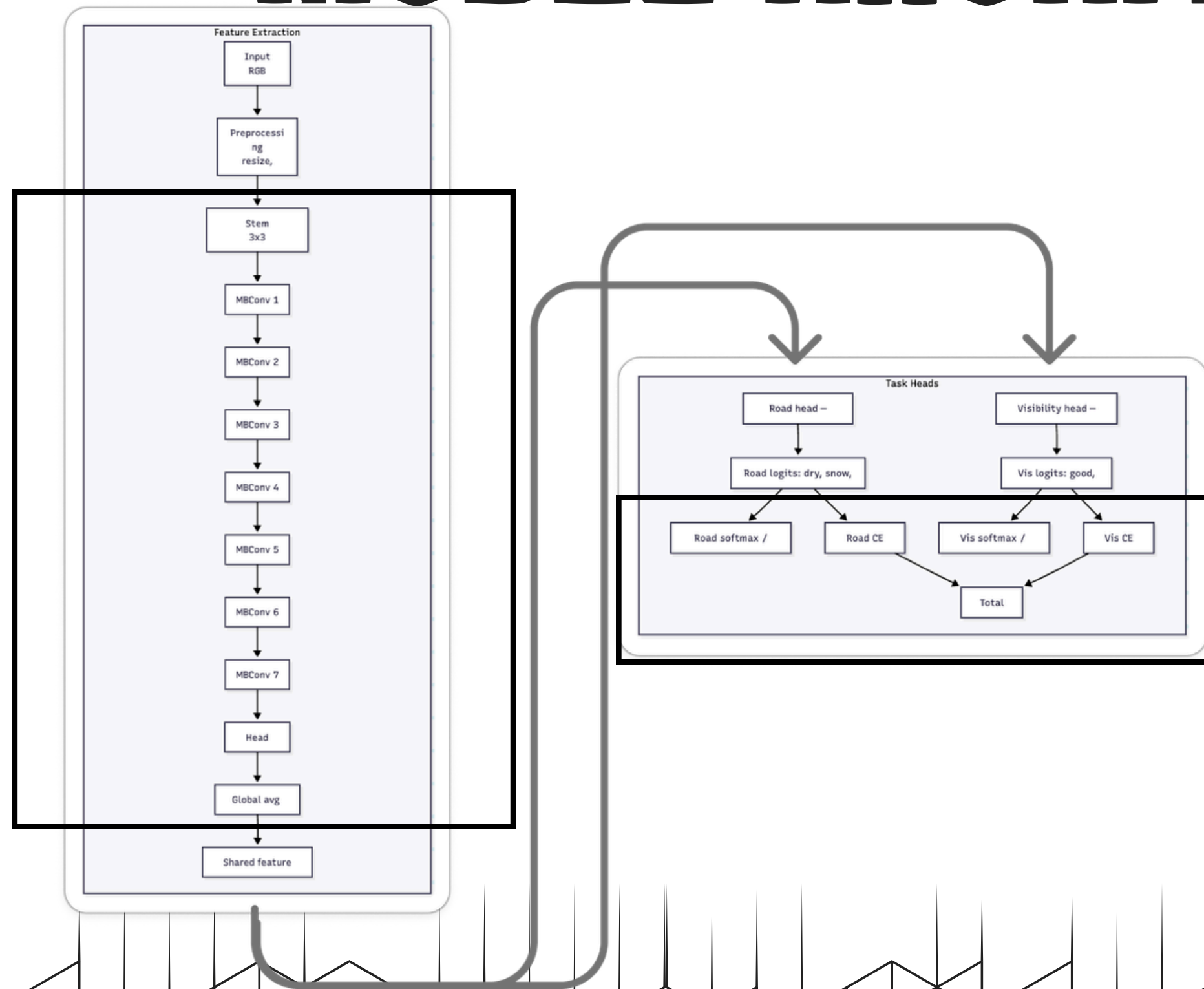
DATASET ANALYSIS + OPS



MODEL ARCHITECTURE



MODEL ARCHITECTURE

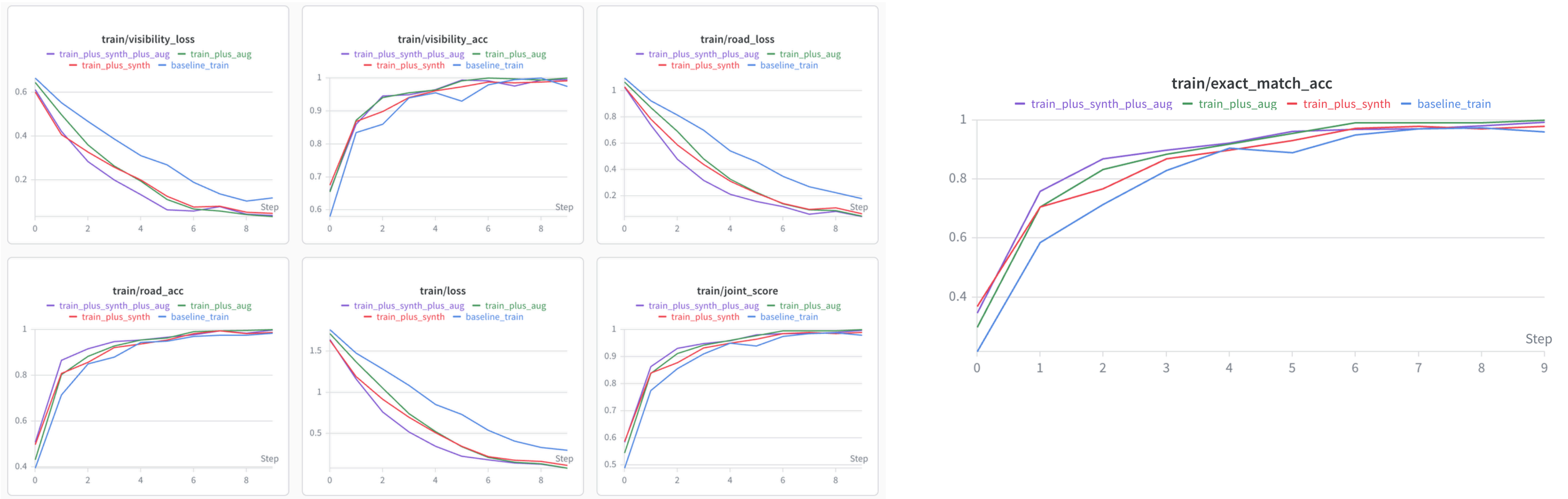


- fast training and inference
- relatively small parameter count
- suitable for our small dataset

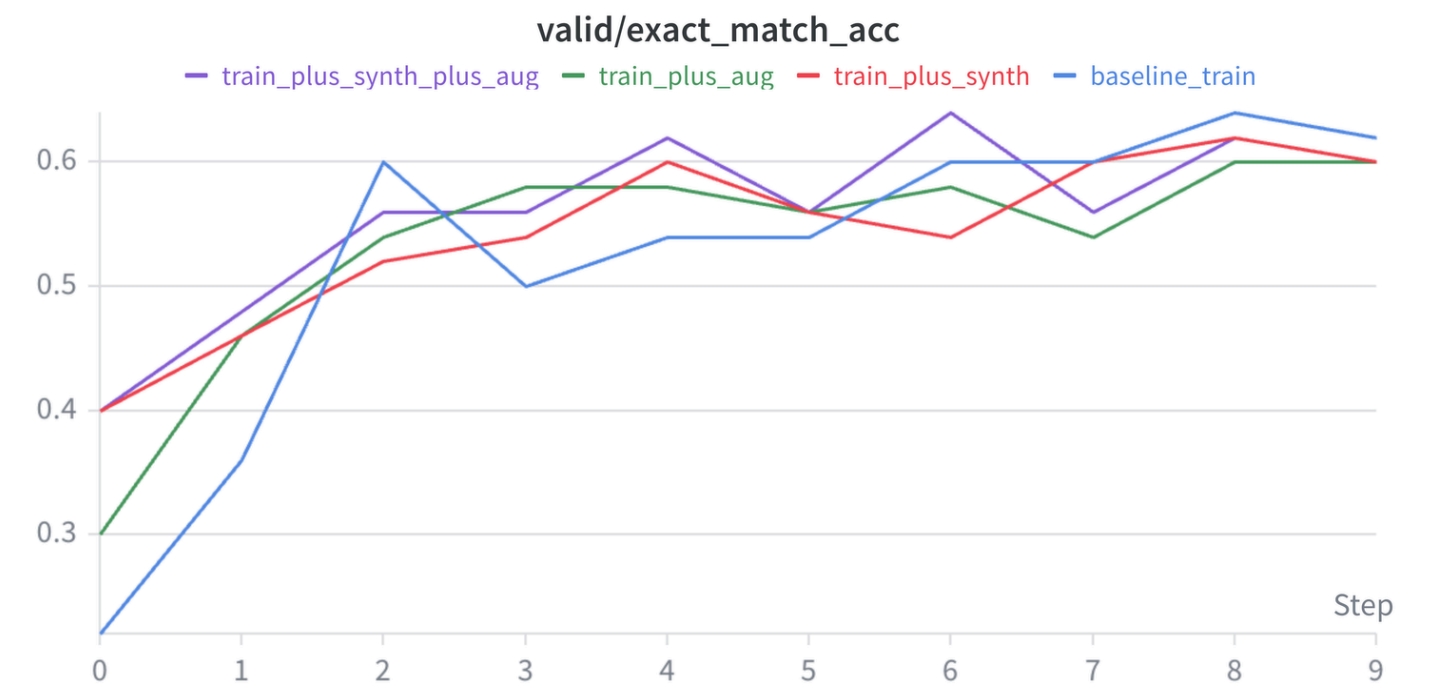
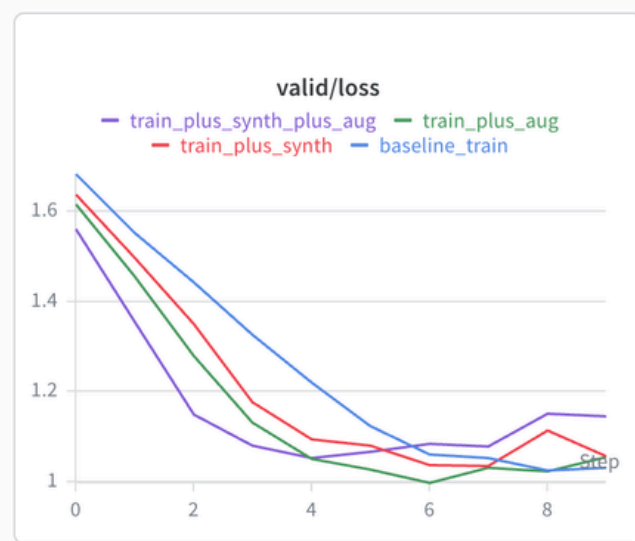
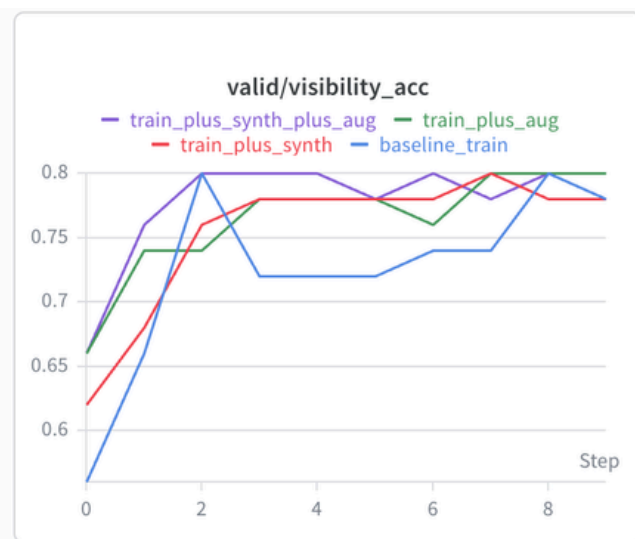
METRICS

- Visibility Cross Entropy Loss (Vis CE)
 - Road Condition CE (Road CE)
 - Total Loss = Vis CE + Road CE
-
- Visibility Accuracy
 - Road Condition Accuracy
 - Joint Accuracy (0.5 for 1 correct label)
 - Exact Accuracy (0 for 1 wrong label)

EVALUATION

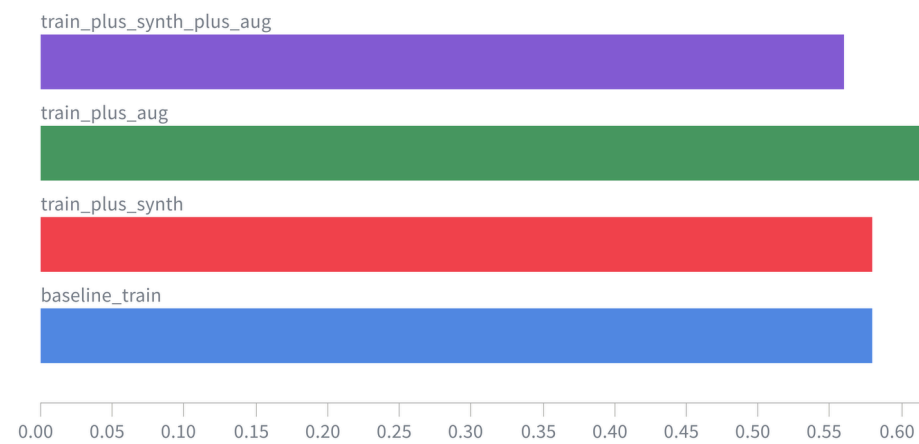


EVALUATION

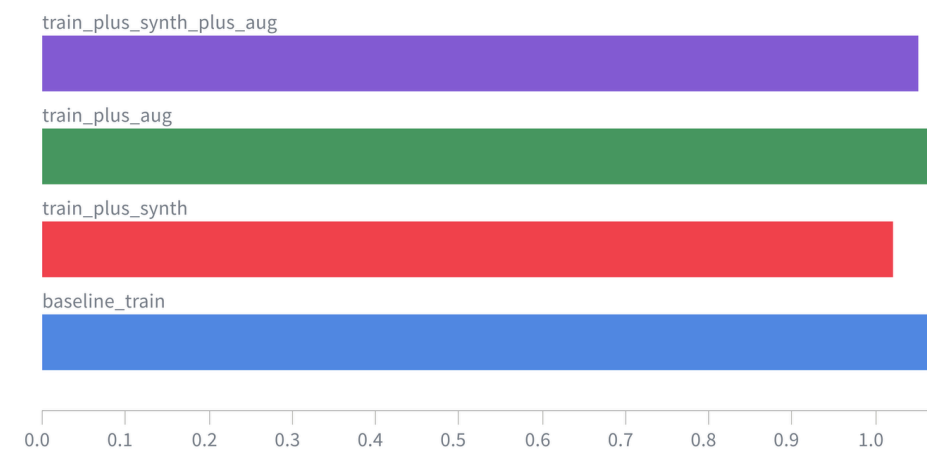


EVALUATION

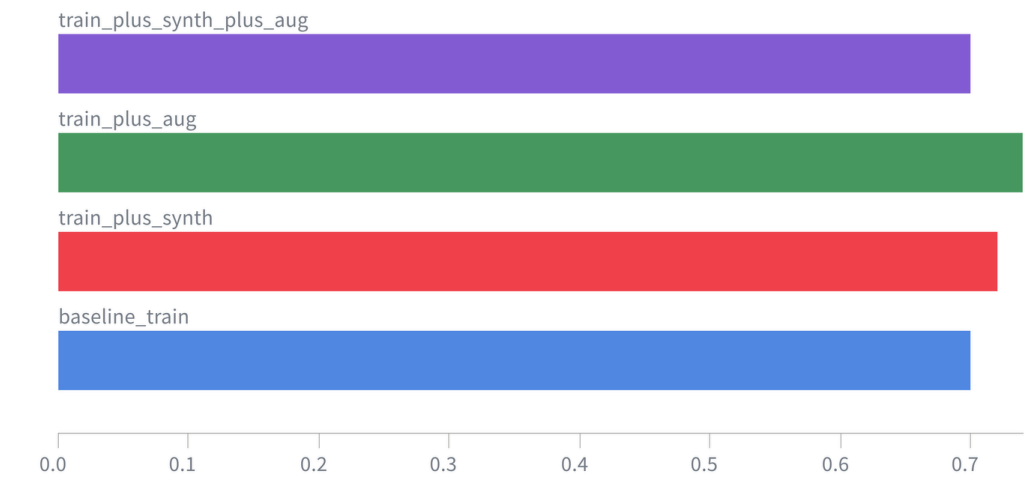
test/exact_match_acc



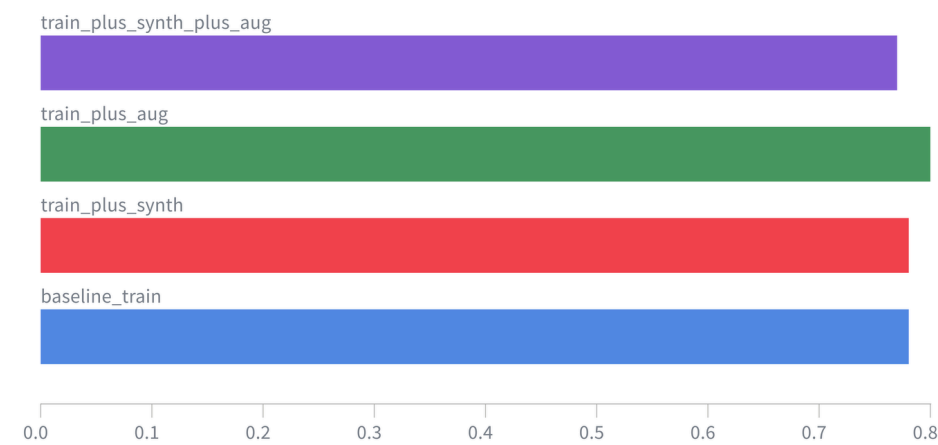
test/loss



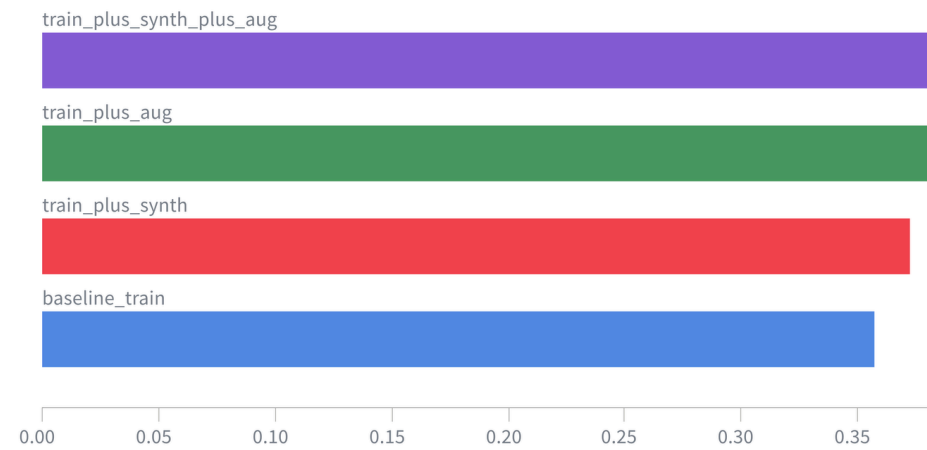
test/road_acc



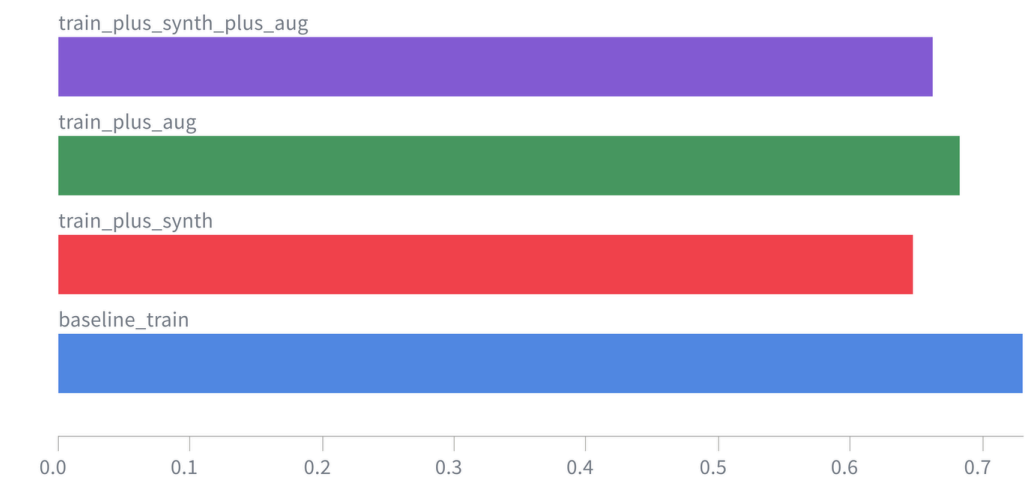
test/joint_score



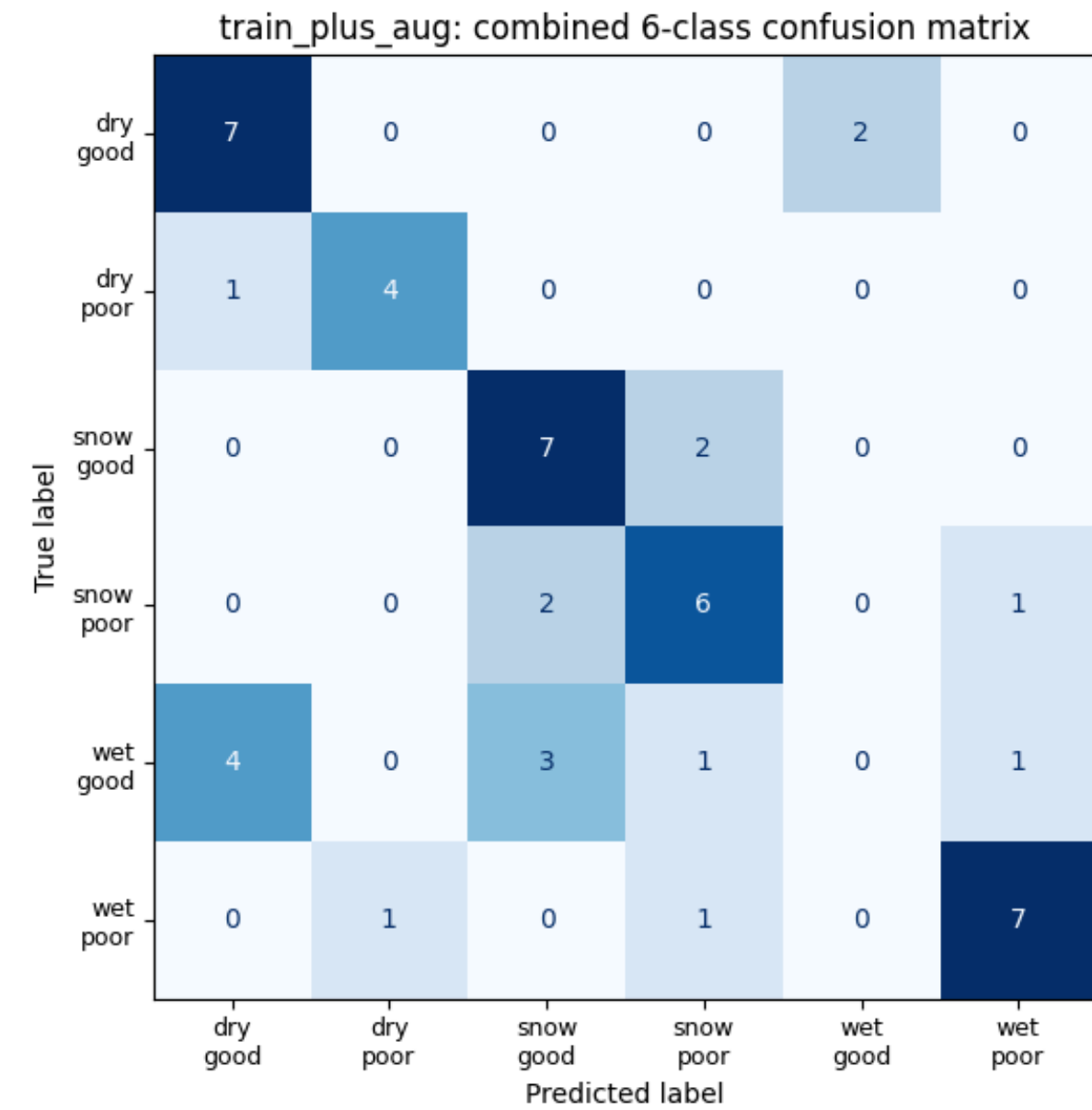
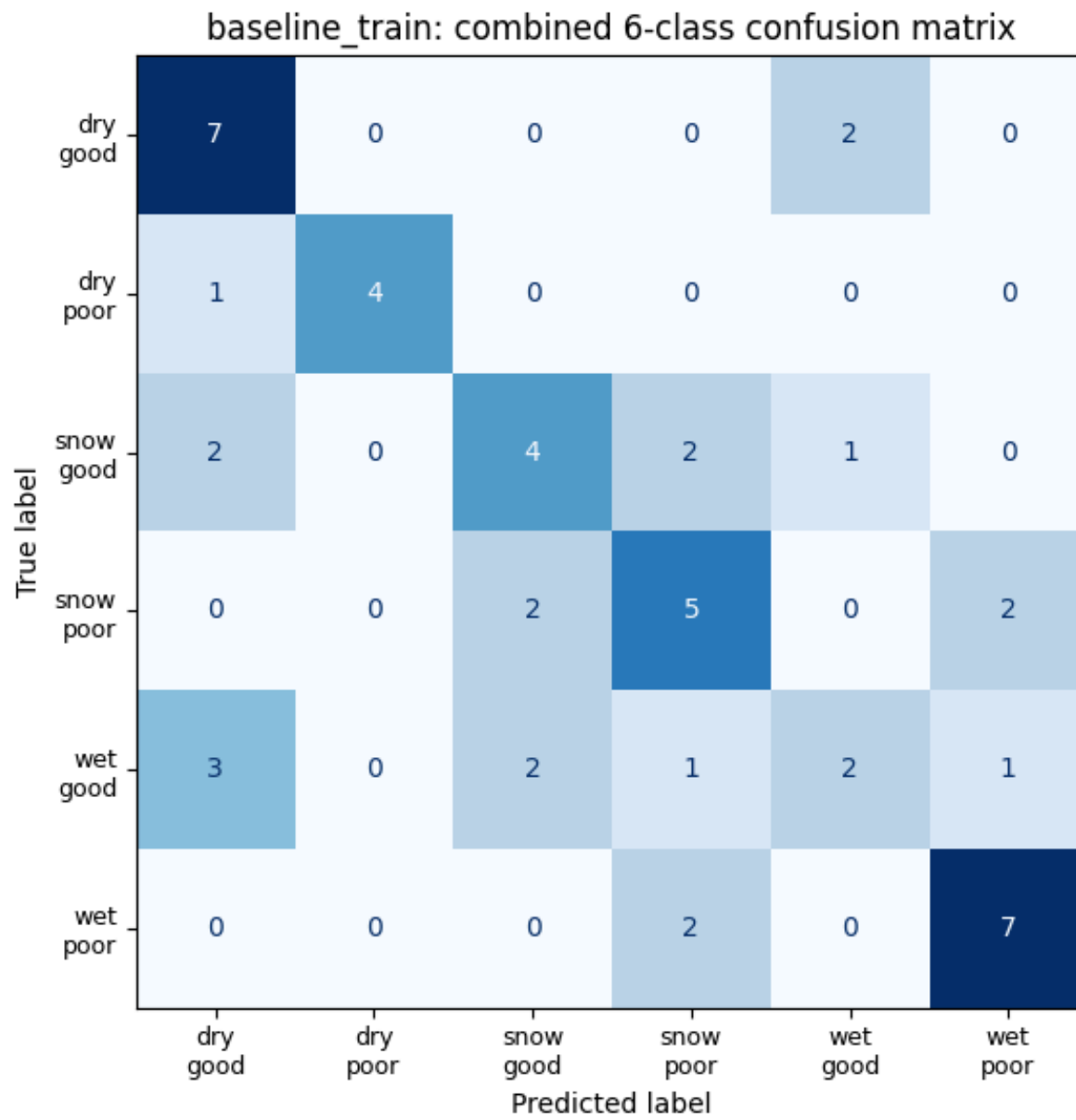
test/visibility_loss



test/road_loss

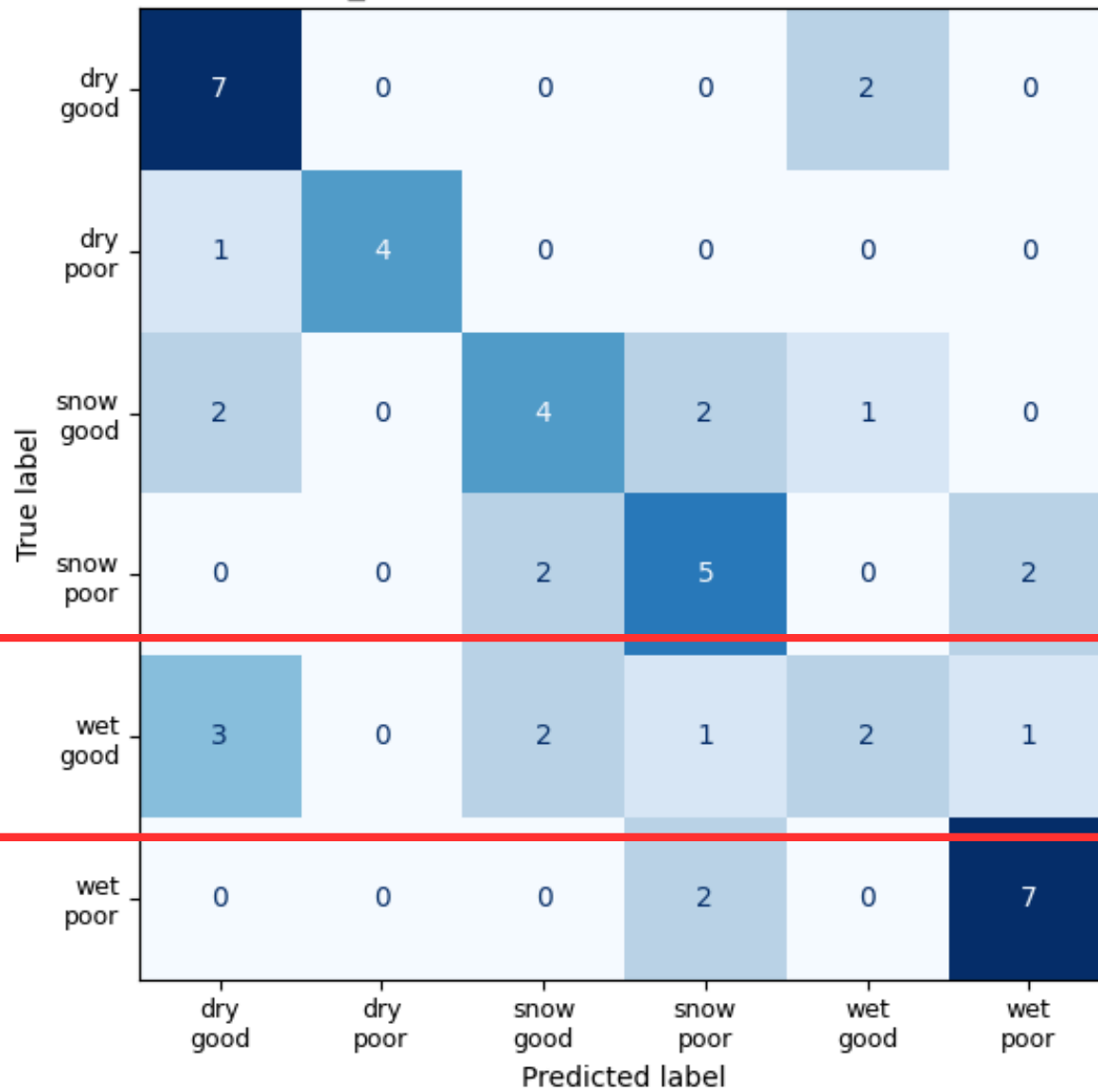


EVALUATION

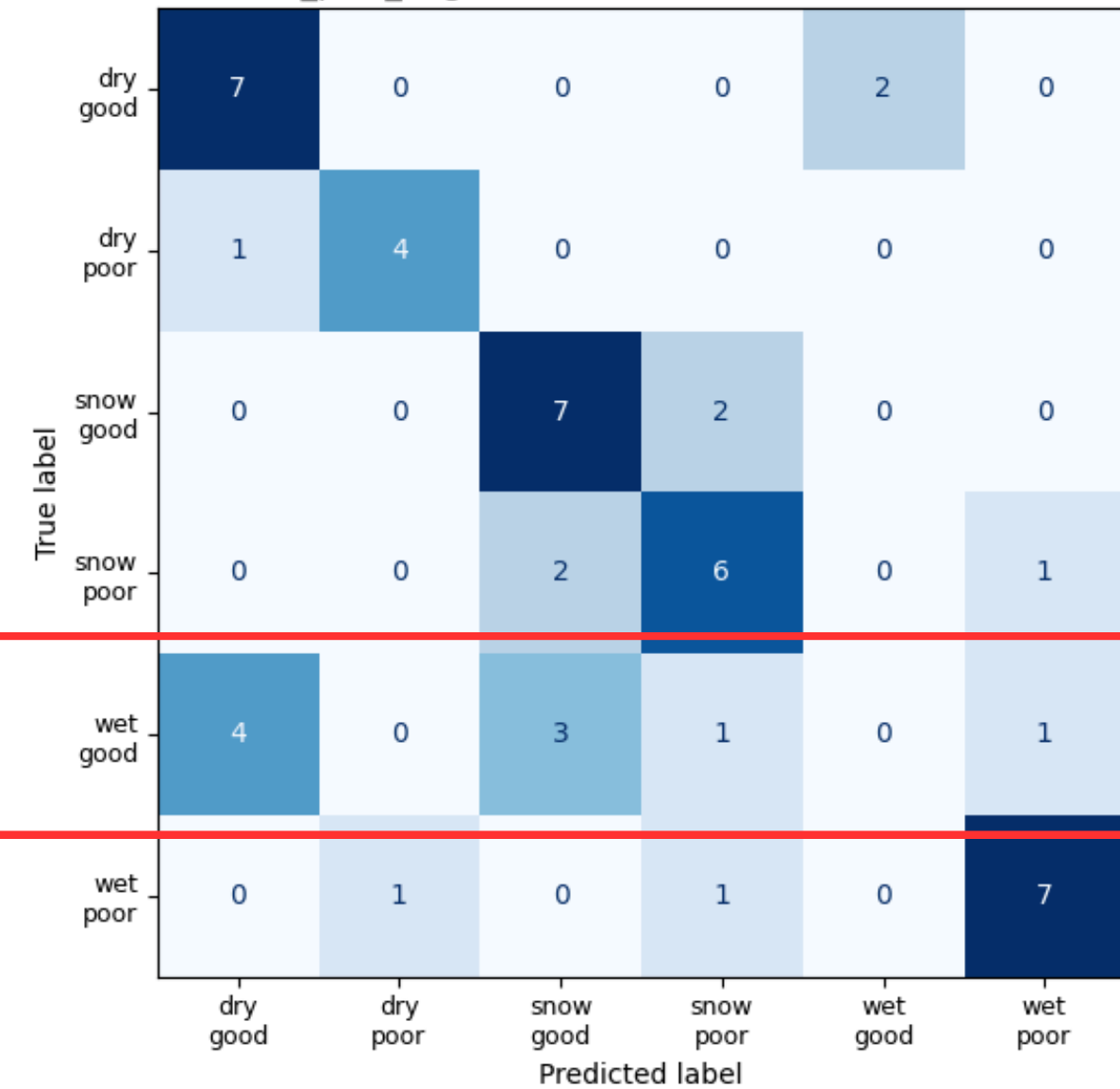


ANALYSIS

baseline_train: combined 6-class confusion matrix



train_plus_aug: combined 6-class confusion matrix



ANALYSIS

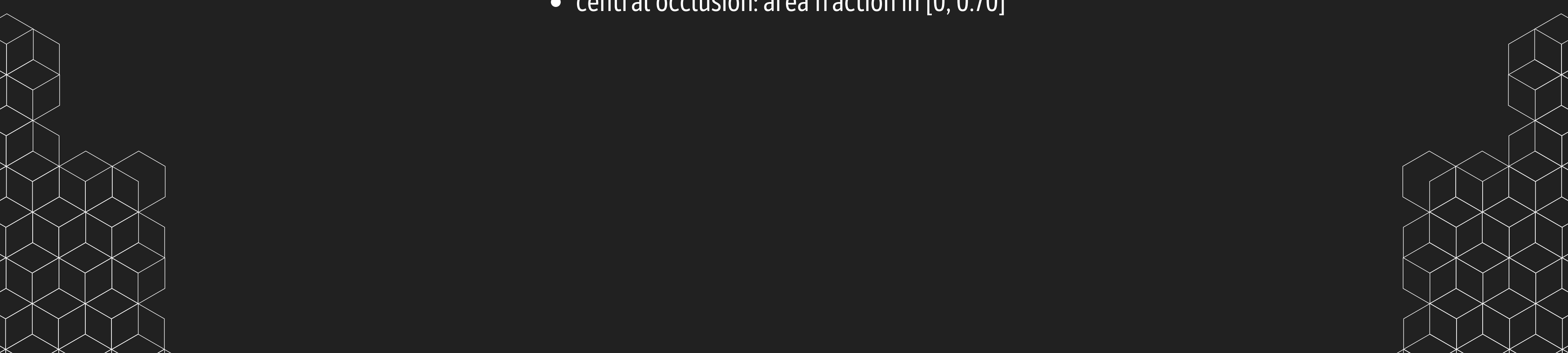


Difficult pictures

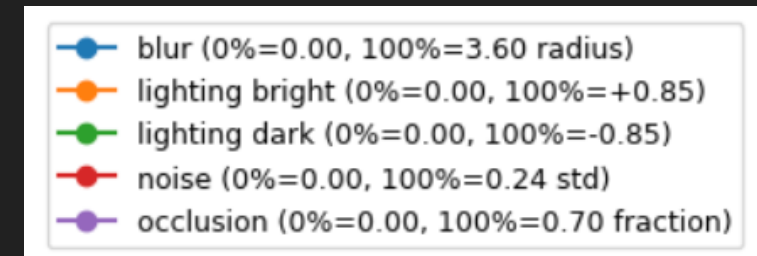
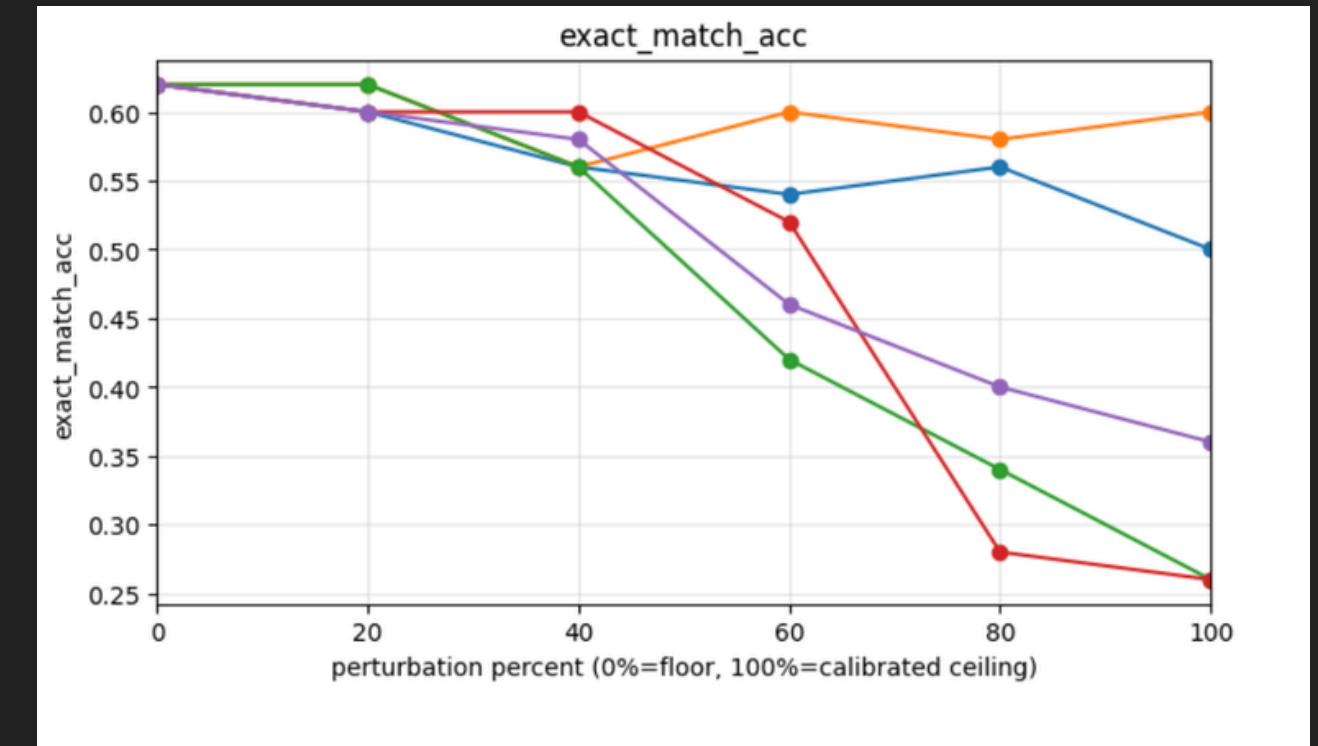
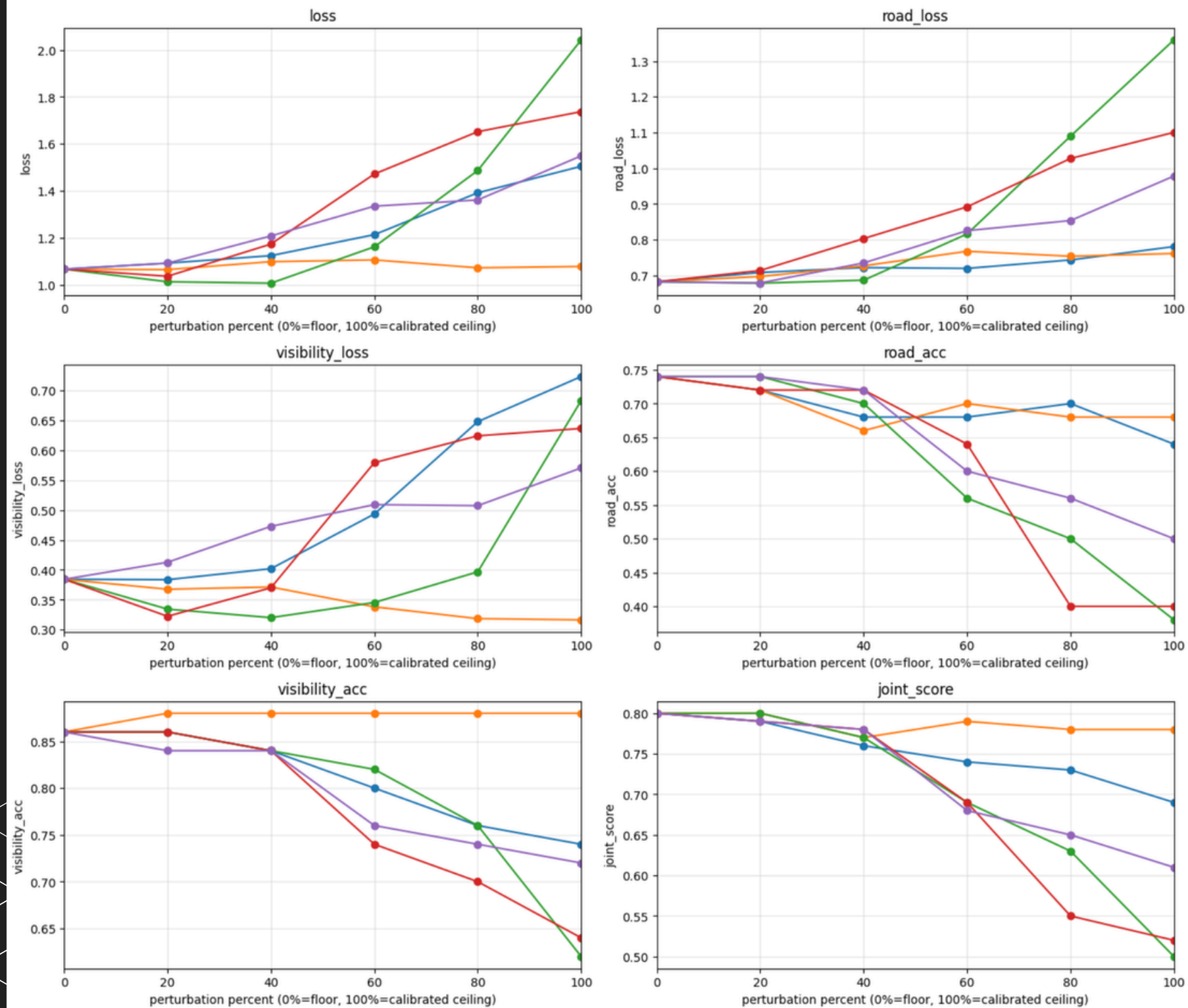


ROBUSTNESS

Perturbations

- Gaussian noise: σ in $[0, 0.24]$
 - Gaussian blur: radius in $[0, 3.6]$ px
 - darkening: lighting_delta in $[0, -0.85]$
 - brightening: lighting_delta in $[0, +0.85]$
 - central occlusion: area fraction in $[0, 0.70]$
- 

ROBUSTNESS



CONCLUSIONS

Contrary to conventional guesses, aug+synth was not the best performing model. Here are my best guesses why:

- GPT Image's resolution was different, which resulted in different features after preprocessing
- The images may have looked too different from the rest of the dataset, which means that the training set might have been pushed OOD from the test set.

It's worth noting that in the validation set aug+synth did perform the best, but the test set represents the open world performance which has been given precedence.



ChatGPT **Prompts:**

- write a gradio app where i can click any test image from a gallery, see the image on one side and the predicted road/visibility classes on the other
- make the app load a hardcoded BEST_MODEL_PATH and show the checkpoint name, corpus name, and probabilities for both classifier heads
- add dials below the image for synthesized noise, blur, lighting change, and occlusion
- change the dials to use percentages instead of raw values, where 0 means none and 100 means the calibrated ceiling
- make sure the app shows the test index, filename, and output_path so i can debug mislabeled samples
- split lighting into one signed percentage control so negative is darkening and positive is brightening

<https://www.gradio.app/main/docs/gradio/blocks>

<https://www.gradio.app/docs/gradio/gallery>

<https://docs.pytorch.org/vision/stable/models/efficientnet>

<https://pillow.readthedocs.io/en/stable/reference/ImageEnhance.html>

<https://pillow.readthedocs.io/en/stable/reference/ImageFilter.html>

<https://github.com/enricivi/RWVC-BDD100K#main-references>

SOURCES

